

Overview of regression methods

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1. Introduction

Regression analysis is arguably the most widely used tool in applied statistics, and has also inspired many important developments in statistical theory. Here we discuss some concepts that can be used to understand several of the most popular approaches to regression analysis. In addition we will review some specific regression methods along with their key properties.

Before proceeding, note that regression itself is somewhat difficult to define in a way that differentiates it from the rest of statistics. In most cases, regression focuses on a conditional distribution, e.g. the conditional distribution of a variable y given another variable x . The symbols y and x may denote either scalars or vectors. Any analysis focusing on such a conditional distribution can be seen as a form of regression analysis.

The variables y and x are referred to by numerous equivalent names. The variable y is known as the “outcome”, the “response variable”, and the “dependent variable”, while x may be referred to as a “covariate”, a “predictor”, a “regressor”, an “independent variable”, or an “explanatory variable”.

2. Major concepts

- *Prediction*: A major use case for regression is to predict (or “forecast”) unobserved values of the dependent variable y when the explanatory variables x are observed. However not all applications of regression analysis focus on prediction. In basic research we may use regression analysis to understand whether and how two quantities are related even when there is no practical role for prediction. Moreover, in many research domains, the strength of the relationship between two variables may be too weak to be useful for prediction, nevertheless regression can suggest important, potentially causal relationships. This use of regression analysis aims to provide insight into mechanisms behind the relationship of x with y .
- *Mean regression*: this term refers to any regression analysis where the population target is the conditional mean function $E[y|x]$; also known as *mean structure modeling*.
- *Variance regression*: this is a class of approaches that model the conditional variance $\text{var}[y|x]$ alone, or that model the conditional variance along with the conditional mean. A common technique is to model $\log \text{var}[y|x]$ since variance is a non-negative quantity. This area is also known as *variance structure modeling*.
- *Quantile regression*: this refers to any method that models a conditional quantile at a given *probability point* in terms of covariates. A common form of quantile regression is *median regression*, which is sometimes favored over approaches targeting the conditional mean because it is more “robust”. Quantile regression using probability points other than 0.5 can be used to understand the skewness, kurtosis, or tails of the conditional distribution of y given x .
- *Single index models*: a single index model is any regression model that is expressed in terms of one *linear predictor* involving the covariates, where a linear predictor has the form $b_1x_1 + \dots + b_px_p$, with the x_j being the observed covariates and the b_j being unknown coefficients

(parameters). In a single index model, all the information in x about the conditional distribution of y given x is contained in this single linear combination of the components of x .

- *Linear model*: Depending on the context, this term can mean any of the following: (i) the conditional mean is a linear function of the covariates, (ii) the conditional mean is linear in the parameters, (iii) the fitted values and/or parameter estimates are linear in the observed values of the response variable, or (iv) the *score equations* satisfied by the parameter estimates are linear. In common use, “linear model” refers to a linear model for the mean structure, e.g. $E[y|x] = \alpha + \beta x$, that is usually fit to the data using *linear least squares*. This model always has properties (ii) and (iii). It may not have property (i), since $E[y|x] = \alpha + \beta x + \gamma x^2$ can also be considered to be a linear model with properties (ii) and (iii) but does not have property (i). It is incorrect to assert that linear models must have property (i), and as a result cannot represent regression relationships between y and x that are nonlinear. An example of a model that does not have properties (i), (ii), or (iii) is the mean structure model $E[y|x] = \exp(\alpha + \beta x)$. If (iv) holds, then the randomness in y transfers linearly to $\hat{\beta}$, which makes many inference tasks especially straightforward.
- *Regression for independent observations*: Most basic regression methods are directly applicable to samples of independent observations (where “independence” in this context means statistical independence conditioning on the covariate values if they are viewed as random). These basic methods may or may not give meaningful results when the observations are statistically dependent, i.e. if $y_i \not\perp y_j \mid x_i, x_j$. More advanced regression methods have been devised for use when the observations are statistically dependent.
- *Heteroscedasticity*: If the conditional variance $\text{Var}[y|x]$ is constant (i.e. does not depend on x), then the conditional distribution of y given x is homoscedastic, otherwise it is heteroscedastic (alternative terminology is “constant variance” and “nonconstant variance”). Heteroscedasticity can be accommodated by some regression procedures (e.g. Poisson regression works best if the mean and variance are equal, which is a particular form of heteroscedasticity). Some regression procedures, like ordinary least squares, work best when the population is homoscedastic, but can still give meaningful results (usually with some loss of power) if the population is heteroscedastic.
- *Mean/variance relationship*: If the variance of a distribution is a function of the mean, there is a mean/variance relationship. For example, in the Poisson distribution, the variance is equal to the mean, and in the negative binomial distribution the variance has the form $\mu + \alpha\mu^2$, where $\alpha \geq 0$ is a “shape parameter”. However in the Gaussian distribution, the variance is unrelated to the mean (i.e. it is a constant function of the mean) so there is no mean/variance relationship (it is *homoscedastic*).
- *Overdispersion/underdispersion*: these terms refer to misspecification of a working regression model relative to the population. If the true conditional variance is greater than the conditional variance of the model being fit to the data, then there is overdispersion. If the true conditional variance is less than the conditional variance of the model being fit to the data there is underdispersion.
- *Repeated measures*: This is one reason that data may be non-independent. Repeated measures (or “clustering”) refers to any setting in which the data fall into groups, and the

observations in any one group are more similar to each other than they are to observations in other groups. This can happen, for example, due to unobserved covariates that are stable (constant) within a group. Repeated measures usually arise by design, e.g. if we are working with longitudinal data or a cluster sample.

- *Marginal regression*: This is a form of regression analysis where the estimation target is the marginal regression function $E[y|x]$, even though the data may be clustered or otherwise dependent. Some methods for marginal regression also give insight into the marginal variance function $\text{Var}[y|x]$ and marginal covariances $\text{Cov}[y_1, y_2|x_1, x_2]$. If the data are dependent, a “complete” model should target the joint distribution $P(y_1, \dots, y_n \mid x_1, \dots, x_n)$. A marginal model allows us to focus on simpler “marginal” relationships.
- *Multilevel regression*: This is an alternative term for “random effects modeling”. It emphasizes the fact that in many data sets, there are complex inter-relationships between the observations that are not explained by the covariates. These inter-relationships allow us to speak in terms of unobserved “random effects” that are incorporated into the linear predictors of one or more observations. The random effects give rise to dependence, and also, in nonlinear models, they give rise to different ways of defining a “regression effect”. Multilevel models also provide a way to model variances and covariances, modeling them through random effects rather than directly (as in a marginal model).
- *Conditional/marginal effect* (in multilevel regression): In a multilevel model, a “marginal effect” is usually defined as the change in $E[y|x]$ corresponding to a one unit change of a specific covariate x_k . A “conditional effect” is usually defined as the change in $E[y|x, u]$ for a one unit change in x_k , where u is an unobserved random effect. For linear models, conditional and marginal effects are the same, but in nonlinear models the two types of effects differ. Methods for nonlinear regression target either the marginal effects, or the conditional effects, but usually not both. In most cases the conditional effect will be numerically larger than the marginal effect. Note that the word “effect”, while widely used, conveys causality that may not be warranted.
- *Conditional/marginal effect* (in single-level regression): Another use of the term “marginal effect” arises in single-level regression models. In this case the marginal effect is the change in $E[y|x_k]$ corresponding to a one unit change of x_k , while the conditional effect is the change in $E[y \mid x_1, \dots, x_p]$ corresponding to a single unit change in x_k , with the other variables x_j for $j \leq k$ held fixed. When referring to this type of marginal effect, the marginal and conditional effects generally differ even in a linear model.
- *Nonparametric regression*: There isn’t always a bright line between “parametric” and “non-parametric” regression methods, but in general, parametric methods are less “flexible” and may only work well if certain strong conditions on the population hold. Nonparametric methods may work well in a broader class of population settings, and are more robust to difficult to test assumptions being violated, but often have lower power and precision.

3. Generative models versus models for specific features of the conditional distribution

Regression analysis usually centers on the conditional distribution of the response given the predictors, which can be denoted $P(Y|X)$. In practice, we can only aim to recover specific features of this conditional distribution such as the conditional mean $E[Y|X]$, the conditional variance $\text{Var}[Y|X]$, or conditional quantiles $Q_{p[Y|X]}$. While it is common to discuss regression in the setting of generative models such as $Y = X^T \beta + \varepsilon$ this is both too ambitious and quite limiting. Writing this expression implies that we need to assume a strong model for the conditional distribution (in this case an additive model that implies homoscedasticity), even when our goal is to estimate specific features such as the conditional mean (which, when estimated using least squares does not require homoscedasticity to hold).

Important regression procedures such as generalized linear modeling and the Cox proportional hazards model cannot easily be expressed in the additive generative form. Moreover, focusing on the generative form downplays the important role of robustness to failure of certain less important aspects of the generative model. For example, as we will see when we discuss GLMs, the so-called Poisson regression model does not require the distribution of Y given X to be Poisson, but instead can apply to any setting where $\log E[Y|X = x] = \beta^T x$ and $\text{Var}[Y|X] = E[Y|X]$ (and this equality can easily be relaxed to a proportionality).

4. Models, fitting procedures, and algorithms

It is important to distinguish between the various regression model structures (e.g. different model parameterizations), and different methods for fitting a regression model to data. For example, the “linear mean structure” model is a regression model in which the conditional mean function $E[y|x]$ is expressed as a linear function of the predictors x (i.e. in terms of a linear predictor). There are many “fitting procedures” (algorithms) that enable one to fit this linear model to data, including least squares, penalized least squares, and many variations of robust regression, maximum likelihood regression, and Bayesian regression. However all of these fitting procedures are fitting the same model structure to the data.

As an example, least squares is a fitting procedure that can be used to fit a linear model to data. The least squares fitting procedure has statistical properties (i.e. it is known to be efficient and consistent in many settings). A different (e.g. Bayesian or penalized) procedure for fitting the same class of models will have its own, potentially different statistical properties (e.g. it may be consistent in some settings where least squares is not and vice-versa).

Algorithms are specific numerical procedures used to fit a model to data. For example, we may use iteratively reweighted least squares to fit a generalized linear model to a dataset. In many cases, the algorithm exactly minimizes an explicit loss-function, and therefore the algorithm itself does not impact the statistical properties of the analysis (e.g. we can use the QR or SVD approaches to solving the linear least squares problem and always get the same result unless the model is degenerate). In a few settings, e.g. nearest neighbor regression, regression trees, or deep neural networks, people may say that “the algorithm is the model”. In these settings, there is a mean structure model, but the model is extremely flexible and the properties of the

procedure result mostly from the algorithm rather than from the structural form of the model. An early exploration of these ideas was Leo Breiman's [two cultures](#)^o paper.

5. Some specific regression analysis methods

- *Least squares*: ordinary least squares (OLS) is the most basic type of curve fitting. It is most effective when the conditional mean function is linear in the covariates, and the conditional variance is constant (i.e. we have “homoscedasticity”). Both of these restrictions can be worked around – nonlinearity of the mean function can be accommodated using basis functions, and heteroscedasticity can be accommodated using inverse variance weights (in which case we are doing “weighted least squares”, or WLS). Also, heteroscedasticity only impacts statistical efficiency, which may not be a major concern when fitting simple models to large datasets.
- *Generalized Linear Models (GLM)*: GLM's are an extension of linear models that introduce *link functions* and *mean/variance relationships*. The link function allows the expected value of the response variable to be expressed as a known transformation of the linear predictor. The mean/variance relationship expresses how the conditional variance of the response given the predictors relates to the conditional mean of the response given the predictors. GLM's are often (but not always) a better alternative to using linear least squares with a transformed predictor (e.g. instead of regressing $\log y$ on x using a linear model, regress y on x using a GLM with a log link function). GLMs can be used as a likelihood-based approach, but there is also a quasi-likelihood theory for GLMs that justifies their use in much broader settings, e.g. we can use Poisson regression if the data are not integer-valued, or do not follow a Poisson distribution, or even if the mean/variance relationship is violated.
- *Generalized Estimating Equations (GEE)*: GEE is an extension of GLM that allows for certain types of statistical dependencies between the observations. A GEE is determined by specifying the GLM that it is derived from, and a “working model” for the correlation structure. The fitting and inference in a GEE is robust in that the working dependence model can be misspecified, and the estimates and inferences will still be valid (this can be stated in more precise terms but we will not do that here). GEE estimates the “marginal mean structure”. In the linear case, GEE is closely related to the more basic technique of “generalized least squares” (GLS).
- *Multilevel linear models*: multilevel (or mixed) linear models are an extension of the basic linear model in which there are (usually) one or more covariates, and also “random effects” which describe how the observations are correlated with each other. These unobserved random effects can be viewed as missing information that reflects additional structure in the population not captured through the covariates. In the case of a linear mean structure, there is essentially a 1-1 correspondence between mixed (linear) models and linear GLS/GEE models, in that both estimate the same population target (the conditional mean function), but using different estimators. The mixed linear model will give more precise estimates of variance parameters than GLS/GEE when the model is correctly specified, but may be less robust to misspecification of the dependence structure. It is a very rich framework that can be used to account for a variety of structures in the population that are difficult to model in other ways, including clustering, multilevel (nested) clustering, crossed clustering, and

heterogeneous partial associations (e.g. the coefficient for a covariate differs across many known subpopulations).

- *Multilevel GLM's*: these are one of the most challenging classes of regression models, especially from a computational perspective. Structurally, they are very similar to linear mixed models, and in practice, can be interpreted in a similar way, except for the important distinction that in a multilevel GLM, the marginal and conditional mean structures differ (which is not the case for a multilevel linear model).

5.1. Other forms of regression:

- *Survival regression* – this is a large set of techniques used for modeling units that transition among states over time, and more narrowly for handling censored data.
- *Conditional regression* – this is a useful but narrowly applicable “trick” in which by conditioning on certain statistics, a multilevel model is essentially converted into a single-level model. The most familiar forms of this technique are single-level conditional logistic and Poisson regression. In both cases, we can have clustered data (which would more often be handled using mixed effects or GEE), but by conditioning on the observed total of the outcome values within each group, the observations become conditionally independent, and can be rigorously fit using a single-level likelihood approach.
- *Fixed effects regression* – this is an approach for working with clustered data in which “fixed” intercepts (covariates) are introduced indicating each cluster. When the sample size per cluster is sufficiently large, this can be a robust and effective way to accommodate clustering, but with many small clusters, high-dimensional models result and poor efficiency or inconsistency (the *Neyman-Scott problem*) may result.
- *Local regression* – this is a very flexible approach to capturing nonlinear regression relationships. It is an example of a regression method that is not fitting a single-index model; it is generally seen as being limited by the “curse of dimensionality”, so that it cannot be applied with more than a handful of covariates, unless the sample size is very large.
- *Additive regression* – this is a way to restrict the general kernel regression technique to avoid the curse of dimensionality. The conditional mean function $E[y|x]$ is modeled as $g_1(x_1) + \dots + g_p(x_p)$, where the $g_j(\cdot)$ are unknown univariate functions. The model is additive over the covariates, which is a strong restriction, but generalizes classical linear models by allowing each covariate to be transformed in an arbitrary way.
- *Dimension reduction regression* – this is a class of regression approaches that posit a multi-index structure and an unknown link function. Specifically, $E[y|x]$ may be modeled as having the form $g(b_1^T x, \dots, b_k^T x)$, where the b_j are vectors of regression coefficients, and g is an unknown link function. The focus is on estimating the regression “directions” b_j , not on the link function.
- *Generalized method of moments (GMM)* – this is a technique for efficiently estimating the parameters of nonlinear models using only the moments. It is mainly used when it is important to estimate regression effects without requiring a model for the full conditional distribution to be specified.

- *Multivariate regression* – these are techniques for regressing a vector of dependent variables on a vector of independent variables. Information is shared across the regressions to allow them to be fit more accurately as a collection, compared to performing separate regressions to the components of the vector dependent variable.
- *Machine learning/algorithmic regression* – this is a broad and loosely-defined collection of methods for regression analysis in which complex representations of regression functions, e.g. trees, ensembles of trees, or neural networks, are fit to data, usually with substantial regularization. The distinction between “machine learning” and “statistical regression” is generally artificial and not useful to make, but some techniques, especially neural networks, are often viewed as being part of machine learning.
- *Manifold regression* – this is a class of methods that can be applied when the independent and/or dependent variables lie on a manifold that is not a metric space. Since many regression procedures depend on distances, this usually takes place in the setting of Riemannian manifolds. A relatively simpler example of manifold regression is *compositional regression* in which the response lies on a circle.